

Juan José Aguado

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PROFILE

Data Scientist and Machine Learning Engineer specialized in deep learning, computer vision, and LLM-driven analytics. Experienced in designing and deploying scalable AI pipelines using Python, PyTorch, and Azure. Passionate about transforming unstructured data into actionable insights through experimentation, reproducible modeling, and automation.

EDUCATION

Pontificia Universidad Javeriana

B.S. in Systems and Computer Engineering (Data Engineering Focus)

Cali, Colombia

Aug. 2020 – Jun. 2025

Universidad Icesi

Master's in Applied Artificial Intelligence

Cali, Colombia

Jul. 2025 – Dec. 2026 (Expected)

EXPERIENCE

Data Scientist

Atinna

Oct. 2024 – Feb. 2026

Cali, Colombia

- Designed and automated end-to-end social listening pipelines for multilingual data ingestion, LLM model execution, and analytics reporting, improving productivity by **50%**.
- Developed semantic clustering systems using **embeddings** and **unsupervised learning** to identify conversation themes across millions of social media posts.
- Enhanced classification accuracy for **stance and sentiment analysis** through **few-shot learning** and prompt engineering with **DSPy** and **LangChain**.
- Integrated **Langfuse** to monitor and version prompts in production, ensuring traceability and continuous LLM improvement.
- Built scalable backend services in **Python** connected to **Azure SQL**, supporting dynamic **Power BI** dashboards for real-time client insights.
- Optimized collaborative workflows through **Git**-based versioning and reproducible model deployment pipelines, accelerating iteration cycles.
- Architected and deployed end-to-end LLM workflows for web video extraction, AI description inference, and metadata integration into structured analytics.

Research Assistant – Applied Software Engineering Group

Pontificia Universidad Javeriana

Jun. 2024 – Dec. 2024

Cali, Colombia

- Developed and maintained the university's automated email service using **Java** and **Maven**, enhancing scalability and reliability.
- Validated and tested API endpoints using **Postman**, increasing delivery efficiency and system uptime.
- Collaborated on refactoring efforts to improve code modularity and maintainability across microservices.

Database Administrator

Fundación Casa de Colombia

Jan. 2024 – Jun. 2024

Cali, Colombia

- Led the design and deployment of a PostgreSQL database system to centralize educational and healthcare program data.
- Coordinated software development using **Jira** and **Bitbucket**, ensuring agile progress tracking and version control.

AI Model Evaluator – Generative AI Models

Outlier (Tasker Platform)

Jun. 2024 – Present

Remote

- Assessed the performance of multimodal **generative AI models** (text, audio, and image) for instruction-following, safety, and factual accuracy.
- Delivered detailed evaluations in English, enhancing the reliability and compliance of next-generation AI systems.

PROJECTS

- Thesis: Prenatal Cranial Plane Classification using CNNs** | *PyTorch* Jun. – Dec. 2024
- Developed a deep learning model to classify fetal brain MRI images into **transventricular**, **transcerebellum**, and **transthalamic** planes.
 - Compared CNN architectures (**ResNet-18**, **DenseNet**, **YOLOv8**) for performance, achieving over **92% accuracy**.
 - Preprocessed MRI data with normalization and augmentation to enhance model generalization.
 - Validated results against expert medical classifications, demonstrating AI's potential for clinical support.
- Email Intelligence System: LLM Agent Classification with RAG** | *LangGraph, FastAPI, Docker* 2025
- Designed and implemented a microservices system for intelligent email classification using **LLM agents** orchestrated with **LangGraph**, featuring confidence-based conditional routing across workflow nodes.
 - Implemented **RAG (Retrieval-Augmented Generation)** with **ChromaDB** and **sentence-transformers** to retrieve semantically similar examples and enhance low-confidence classifications via few-shot learning.
 - Built a **3-microservice architecture** (API Gateway, Classifier, Vector DB) orchestrated with **Docker Compose**, with isolated Docker networks and internal HTTP communication between services.
 - Developed **async** endpoints using **FastAPI** with dependency injection, Pydantic type hints, and structured error handling following production best practices.
 - Designed a **provider-agnostic LLM system** supporting Gemini, OpenAI, and Anthropic through a factory pattern configurable via environment variables — no code changes required to switch providers.
- Smart Farming IoT Platform on AWS** | *Python, AWS IoT Core, Lambda, EC2, PostgreSQL* 2024
- Designed an end-to-end IoT platform simulating soil humidity, NPK, and weather sensors, publishing telemetry to **AWS IoT Core** via **MQTT** with hierarchical topic design following IoT best practices.
 - Built an event-driven ingestion pipeline with **AWS Lambda** and **SNS** for automated threshold alerts, and a **serverless REST API** using AWS Chalice and API Gateway for sensor management.
 - Persisted telemetry data in **PostgreSQL** on EC2, applying event-driven architecture, certificate-based security with IAM, and decoupled pub/sub messaging patterns.
- Statistical Analysis and Data Science Projects** | *R, Markdown* Jan. 2024
- Performed exploratory data analysis and regression modeling on real estate datasets, identifying key price drivers.
 - Implemented **Monte Carlo simulations** and bootstrapping methods to validate estimator efficiency and the Central Limit Theorem.
- UVa Algorithmic Programming Challenges** | *Python* Jan. – Jun. 2023
- Solved 100+ algorithmic problems involving data structures, dynamic programming, and graph traversal.
 - Strengthened analytical problem-solving and computational thinking skills through iterative optimization.

COURSES

- HarvardX CS50** | *Django, HTML, Git, CSS, JavaScript, SQL* Apr. 2024
- Developed a full-stack web application using **Django**, implementing RESTful APIs and SQL data management.
 - Applied secure development practices, testing, and version control with **Git/GitHub**.

SKILLS

Programming: Python, C/C++, SQL, JavaScript, R, Java
Machine Learning / AI: PyTorch, TensorFlow, Scikit-Learn, YOLOv8, ResNet, DenseNet, DSPy, LangChain, OpenAI API
Data Engineering: Azure SQL, PostgreSQL, Power BI, Docker, Git, FastAPI
Developer Tools: VS Code, Jira, Bitbucket, Langfuse, VirtualBox, Google Colab, Eclipse, Maven
Cloud Platforms: Azure, AWS
Languages: Spanish (Native), English (Advanced)